

Connecting the Device via Wi-Fi

When scanning for available Wi-Fi networks, the MX60 system should show up in the list of detected devices.

The SSID of the MX60 system is in the form: "Trimble MX60 (117070101)", where the number within the brackets is the serial number of the MX60 system.

The password to connect to the MX60 Wi-Fi is provided on separate stickers shipped inside the top case of the MX SCAN Control Unit 2. This password is unique for each system and cannot be changed.

See also *TMI Software User Guide - Establishing Network Connection*.

Storage and Transportation

Delivered Items

The MX60 System is delivered in two different boxes:

1. **Box 1.** This shipping box includes:

- The Trimble MX60 Sensor Unit in its storage case. The dimensions of the storage case are (L × W × H): 806 × 716 × 634 mm.
- The *Control-Unit-to-Sensor-Unit* cable, 5 meters, also included in the storage case.

See picture below (storage case open, shipping box not shown; the storage case also includes a hex wrench).

- Tools: hex wrench 5 and hex wrench 6



2. **Box 2:** This shipping box includes the following items:

- Trimble MX SCAN Roof Rack



- Trimble MX SCAN Power Unit



- Trimble MX SCAN Control Unit 2



This unit has no transportation case as such, but once you have fastened the lid onto it, it can safely be transported without the need for a specific transportation case. You will also find the following items inside the Control Unit transportation case (see also previous picture):

- Stickers of the 'MX SCAN Label WIFI Code'
- Keys for Data Storage Disk
- 2 × Wi-Fi Sticks (pre installed in the Control Unit)
- Quick Start Guide
- SSD Drive (installed in the Control Unit)
- *Source-to-Power-Unit* cable
- *Power-Unit-to-Control-Unit* cable.

In box 2, these items are organized as shown on the picture below.



Where:

- 1: Roof Rack
- 2: Cables
- 3: Control Unit
- 4: Power Unit
- 5: 2 × Data Carrier Dock

Box 2 weighs 75 kg and has the following dimensions (L × W × H): 1170 × 770 × 500 mm.

Use Assumptions

Trimble assumes the different components of the MX60 system are used as follows:

- The Roof Rack is permanently installed on the vehicle roof.
- The Power Unit is permanently installed inside the vehicle.
- The Source-to-Power-Unit cable is permanently installed inside the vehicle.
- The Control Unit is temporarily installed in the vehicle just for operation.
- The Sensor Unit is temporarily installed on the Roof Rack and removed for transfer¹. The GAMS Antenna Kit and its cable are temporarily installed and removed for transfer if the Sensor Unit has been removed.
- External instruments (e.g. DMI) as well as the cabling for these instruments are removed for transfer and storage.
- Cables between Power Unit and Control Unit and between Control Unit and Sensor Unit temporarily installed in and on the vehicle for operation and removed for storage and transfer.

Whenever the Trimble MX60 system is not used (storage, transportation or transfer), the Sensor Unit must be stored in its storage case and the Control Unit must be closed by placing its lid back on.

Storage

Storage conditions are provided in the Technical Parameters section of this manual. See [Environmental, page 53](#).

¹"Transfer" refers to that period of time during which the vehicle is driven without operating the MX60 system, although it's still installed in the vehicle (except for the Sensor Unit which should have been removed from the roof).

Additional Accessories

DMI

The “Trimble MX SCAN DMI Kit” T001507 is an additional mechanical assembly group allowing you to use the DMI (Distance Measuring Indicator) functionality with the Trimble MX60 system.

To improve the measurement accuracy of the Trimble MX60 system, this additional sensor can be used (Zero Velocity Update (ZUPT) information for navigation postprocessing).

For more information about the Distance Measuring Indicator, refer to the *Trimble DMI Installation & Operation Manual* or contact your Trimble dealer.



GAMS

The “Trimble MX GAMS Antenna Kit” is a mechanical assembly group allowing you to use the GAMS (GNSS Azimuth Measurement System) functionality with the Trimble MX60 system.

GAMS is an additional GNSS antenna used in the field to speed up the initialization of the navigation system, hence of the MX60 system. Thanks to the second GNSS antenna, the orientation of the car can be estimated far more quickly compared to what can be achieved with a single-antenna solution. For more details, please contact your Trimble dealer.

For more information about the GNSS Azimuth Measurement System, refer to the *Trimble GAMS Antenna Kit Installation & Operation Manual*.



To connect the GAMS kit to the MX60 system, use the **Ant** connector on the MX60 Sensor Unit.

Operation

Safety Check

- Complete a safety check before and after each mission.
- All broken or damaged components should be exchanged immediately.
- All loose screws should be tightened.
- Any screw must be tightened with the correct torque value as mentioned in [Tightening Screws, page 10](#). Be aware of the appropriate procedure to tighten screws.
- Any dirty or wet part should be cleaned or dried.
- Do not start a mission before solving any issue you may have had previously with the system. Not doing this may damage the system permanently.
- The operator must take care to fulfill all requirements and certifications.

Components Checklist

- Check that the MX SCAN Roof Rack is installed correctly:
 - Check that all the screws of the MX SCAN Roof Rack have been tightened.
 - Check that the MX SCAN Roof Rack components show no cracks or deformation.
- Check that the MX60 Sensor is damage-free (no scratches or deformation of material):
 - Check that all camera lenses are clean and not damaged in any way (scratches or marks).
 - Check that the Sensor is in operation position and properly locked.
- Check the Control Unit for damage or any broken parts.
- Check that the Control Unit is installed correctly and secured.
- Check that the Power is damage-free, and there is no broken parts.
- Check that the Power is installed correctly, air inlet and air outlet are free and the Power is secured.
- Check all cable connections:
 - Check that the connectors are plugged in and fixed.
 - Check that the cables are fixed to the vehicle roof rack before being led inside the vehicle cabin.
 - Check that no cable is damaged or might get damaged during operation.
- Check that the user interface device (laptop, tablet, etc.) is operational.

External Reference Point

The MX60 system needs to know how it has been mounted on the vehicle and how the external components have been mounted relative to the MX60. These mounting measurements are listed below:

- **Installation Height:** Height of the MX60 system above the road surface
- **DMI Lever Arm:** Three-axis distance between the MX60 system and the DMI (wheel sensor)
- **GAMS Lever Arm:** Three-axis distance between the MX60 system and the secondary GNSS antenna.

Measuring Installation Parameters and Lever Arms

All these measurements are performed from the MX60 system's External Reference Point to the corresponding counterpart. The **External Reference Point** is located on the right side, at the back of the MX SCAN Roof Rack.

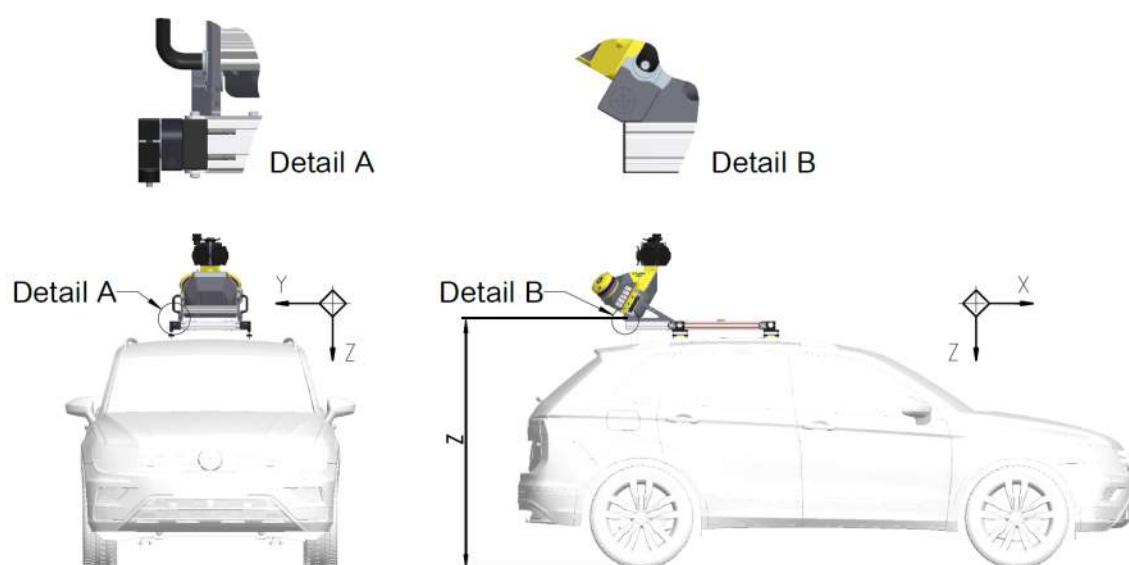


The **Installation Height** is measured from the External Reference Point down to the road surface. The value is always given in meters as an absolute value.

Lever Arm measurements for DMI and GAMS are performed in the so-called Vehicle Frame from the External Reference Point to the center of the corresponding sensor (DMI or GAMS) as measurements of distance (also in meters) along three different axes. The orientation of the axis in the vehicle frame is defined as follows:

Positive X-axis	In forward driving direction
Positive Y-axis	Right side of vehicle
Positive Z-axis	Downward

CAUTION – Make sure you are using the correct sign when measuring Lever Arm values. For example, a DMI mounted on the left wheel has a negative Y Lever Arm value.



Additional Note for DMI

The DMI Lever Arm is a three-dimensional vector defining the offset from the External Reference Point to the sensing center of the DMI. This offset vector is measured in the Vehicle Frame. Lever distances are measured from the External Reference Point to the center of the tread (where the DMI-equipped wheel makes contact with the road). The DMI-equipped wheel must be a non-steering wheel.

Additional Note for GAMS

The GAMS Lever Arm is a three-dimensional vector defining the offset from the External Reference Point to the GAMS Antenna. This offset vector is measured in the Vehicle Frame. Lever distances are measured from the External Reference Point to the Antenna Phase Center of the GAMS Antenna (secondary GNSS Antenna).

Configuring the MX60 system for Operation

All parameters mentioned above need to be measured after having installed the MX60 system and all the additional sensors (DMI, GAMS) on the vehicle. In the next step, the measured numerical values need to be stored as the Vehicle Preset in the TMI Software, which is used to configure the system before a mission is started.

CAUTION – Using incorrect or poor measurement values in the Vehicle Preset may result in faulty navigation solutions. Make sure that all values are entered correctly (on the order of 1 cm). Later on, make sure you select the correct Vehicle Preset before starting a mission. The MX SCAN Roof Rack must be horizontally aligned during the measurement of the height.

Additional Note on POSPac Processing

The origin of the MX60 System Reference Frame is different from the External Reference Point. For this reason, the TMI Software adds additional internal vectors to the Lever Arm values provided in a Vehicle Preset. In POSPac processing, you will be able to read the corrected Lever Arm values and you will see that they are different from the mechanical measurements.

Prerequisites to System Setup and Operation

CAUTION – Before starting, please make sure all connections are secure and the recording SSDs are inserted and locked.

Before starting data capture, make sure the measurement parameters (e.g. sensor setup on the vehicle and capture parameters) have been set properly. The parameters you have defined may be saved as presets under user-defined names and can be selected when starting a mission later.

A standard operation workflow and an “In-the-field operation checklist” are described more in detail in the *Trimble MX60 Quick Start Guide*.

WARNING – The first time you use the Trimble MX60 system, and then every time you start a data collection campaign in a different country, you must specify the name of this country when setting the Wi-Fi access point. This is a mandatory prerequisite. That way, you are sure to comply with the regulations of this country in terms of Wi-Fi use. To set the country of use, run the TMI Software, then in the menu, select **System Administration / WiFi / MX60 WiFi Access Point** and then choose a country.

System Software

The software running the MX60 system is called TMI. It is described in a separate manual, *Trimble TMI Software for Trimble Mobile Mapping Systems User Guide* available from geospatial.trimble.com/en/links?dcs=Collection-129953.

Maintenance and Support

Maintenance

Dismantling or attempting to have a MX60 system repaired by unauthorized personnel can be hazardous and costly. Maintenance should be limited to the cleaning and inspection of external surfaces, lens glass, operating controls, etc.

Preventive Maintenance

The preventative maintenance to be carried out by the operator should include:

- During use:
 - Avoid operating the system in rainy or misty weather.
 - Avoid mechanical shock.
 - Clean and dry the equipment before, during and after use if necessary.
- Before storage:
 - Dry the system thoroughly before storing.
 - Correctly pack the equipment in the transportation / storage case provided.
 - Ensure the transportation / storage case is kept dry and clean inside.
 - Ensure that your equipment is dry before storage.
 - Store within the equipment environmental temperature and humidity limits.
- During transportation:
 - Correctly pack the equipment in the supplied transportation cases.
 - Do not allow the equipment to slide around inside transport vehicles or containers.
- General:
 - Carry out regular functional testing of the system.
 - Detect and report damage, malfunctions and poor performance.

Lubrication of Movable Parts

The lock bars as well as the fast lock mechanism of the Roof Rack should be cleaned and re-lubricated frequently. After cleaning, simply use a spray oil for re-lubrication.

Cleaning the Camera Lens Glass

- Perform all cleaning operations in a clean environment.
- To remove dust and dirt, first try to gently blow the debris off with an air compressor.
- If the debris will not come off using an air compressor, apply a small amount of optics cleaner (e.g. Photographic Solutions Eclipse Optics Cleaner) or ethyl alcohol to a clean lens cloth (e.g. Pentax lens cloth). The cloth should be moist but not dripping.
- Wipe the cloth or swab along the length of the glass surface in smooth movements. Do not press hard on the surface or rub repeatedly on one spot.
- If pooling or streaks occur, there may be too much solution—wait for it to dry, then repeat.
- When finished cleaning, examine the glass surface in light.
- If dust spots remain, repeat this procedure using a clean lens cloth.

Support

General

If you need support for the Trimble MX60 or Trimble Mobile Mapping software, please contact our Customer Support team via email:

Trimble Support Email **mx_support@trimble.com**

Or call:

Support Americas: +1-289-695-4416

Support APAC: +86-105-603-4179

Support Europe & Rest of World: +49-7351-474-0237

Please report your support case as precisely as possible. Please provide:

- a short description of the problem.
- the workflow you used and how to reproduce the problem.
- If occurring during a mission:
 - the location of the mission.
 - environmental conditions.

Also send the following information about your Trimble MX60 system:

- System log file.
- Serial number.
- Total operating hours of your Trimble MX60 system since purchase.
- Photos or videos may also be useful to describe the problem.

To register your product or check the status of your system maintenance, please register your system and software at <https://mytrimbleprotected.com/>.

My Trimble Protected enables Trimble users, channel partners, and end customers to streamline business processes for customer engagement and service. As a centralized instrument panel, My Trimble Protected manages registrations, serial number lookups, product catalogs, reports, settings, and locator processes. It allows users to configure, monitor, and optimize each aspect of the process.

For terms and conditions, see www.trimble.com/Support/Terms_of_Sale.aspx.

Downloads

Documentation for the Trimble MX60 system can be found at <https://geospatial.trimble.com/en/products/hardware/trimble-mx60>.

Appendix

Technical Specifications

MX60 System

Power Requirements

Parameter	Value		
Input Voltage Range:			
MX60 System (Core and Pro)	12 V to 16 V	DC (160 W typical)	
MX60 System (Premium)	12 V to 16 V	DC (170 W typical)	
DC Current: Depends on temperature.	Standby	In Operation	Startup
MX60 System (all configurations)	9 A	12 A	25 A @ 12.8 V

Weight

Parameter	Value
MX60 Sensor Unit (Core)	24 kg
MX60 Sensor Unit (Pro)	26 kg
MX60 Sensor Unit (Premium)	28 kg
MX SCAN Control Unit 2	13 kg
MX SCAN Power Unit	9 kg
MX SCAN Roof Rack	18 kg
Source-to-Power-Unit Cable, 5 meters	2 kg approx.
Power-Unit-to-Control Unit Cable, 3 meters	1.5 kg approx.
Control-Unit-to-Sensor-Unit Cable, 5 meters, STD	3.5 kg approx.

Environmental

Parameter	Value
Operating Temperature Range ¹	-10 °C to +50 °C (14 °F to 122 °F)
Storage Temperature Range	-20 °C to +50 °C (-4 °F to 122 °F)
Operating Relative Humidity Range ²	20 % to 80 %
Storage Relative Humidity Range	20 % to 95 %
IP Rating	IP64 (Sensor Unit) IP30 (Power Unit and Control Unit)
Maximum Vehicle Speed (with operating or non-operating system on board)	110 km/h (68 mph)
Recommended Maximum Vehicle Speed (with operating system on board)	80 km/h (50 mph)

Spherical Camera Performance

The spherical camera system included in the Trimble MX60 Sensor Unit consists of six single cameras that deliver a spherical image. The performance data of the two different cameras, as applied in the MX60, are given below.

Parameter	Value - MX60 Core	Value - MX60 Pro and Premium
Image resolution (total)	30 megapixels	72 megapixels
Image format	PGR (compressed JPEG)	PGR (compressed JPEG)
Shutter system	Global shutter	Global shutter
Focal length	4.40 mm	6.94 mm
Focus distance	About 2.0 m (sharpness from 0.7m to infinity)	Calibrated from 2.0 m to infinity
Field of view	Approx. 90 % of full sphere	Approx. 90 % of full sphere
Number of single cameras	6	6
Image resolution of single cameras	5 megapixels	12 megapixels
Sensor type	CMOS	CMOS
Sensor size (single camera)	2/3"	1.1"
A/D converter	8 bits	8 bits

¹Not exposed to direct sun and without driving less than 10 km/h (6 mph).

²Non-condensing.

Oblique Camera Performance

The oblique camera included in the MX60 Sensor Unit is a down camera. This camera is included on all 3 MX60 configurations.

The performance data of the camera, as applied in the MX60, is given in the table below.

Parameter	Value (down facing camera)
Image resolution	12 megapixels
Image format	MXIPS (compressed RAW format)
Shutter system	Global shutter
Lens focal length	8.0 mm
Focus distance	About 3.0 m (sharpness from 2.0 m to 9.0 m)
Field of view	H: 82.9°, V: 65.9°
Sensor type	CMOS
Sensor size	1.1 "
A/D converter	8 bits

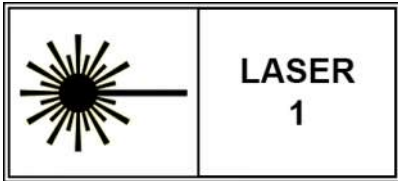
Laser Performance

The Trimble MX60 Sensor Unit includes two scanner units. The scanner consists of a pulsed laser (time-of-flight measurement) and a rotating mirror.

Each of the scanners provides a beam deflection of ~346°.

Scanner and Laser Specifications

Parameter	Value	
Measurement Principle	Time-of-flight measurement:	
Laser Pulse Repetition Rate PRR ¹	500 kHz	1000 kHz
Max. Measurement Range ²	150 m	120 m
Minimum Range	0.6 m	
Range Accuracy ^{3 4}	2 mm	
Precision ^{4 5}	2.5 mm @ 30 m	
Scan Speed (No. of profiles per second)	120 Hz / 200 Hz selectable	
Laser Beam Footprint (Gaussian Beam Definition)	2.6 mm @ exit	

Parameter	Value
Laser Beam Divergence	0.3 mrad (1/e ²)
Laser Wavelength	Near Infrared
Laser Product Classification ⁶ according to IEC60825-1:2014	

¹ Rounded values.

² Typical values for average conditions. Maximum range is specified for flat targets with size in excess of the laser beam diameter, perpendicular angle of incidence, and for atmospheric visibility of 23 km. In bright sunlight, the maximum range is shorter than under an overcast sky.

³ Accuracy is the degree of conformity of a measured quantity to its actual (true) value.

⁴ One sigma range under test conditions.

⁵ Precision, also called reproducibility or repeatability, is the degree to which further measurements show the same result.

⁶ The classification is based upon the assumption that the laser beam is continuously scanned.

AP+ Positioning System Performance

The MX60 mobile mapping system is equipped with different positioning systems depending on the configuration.

Each configuration features a high-performance, precision GNSS receiver and the industry-leading, Applanix® IN-Fusion™ and ProPoint® GNSS-Inertial integration technology running on a dedicated and powerful Inertial Engine (IE) board.

Assuming the best operation conditions regarding satellite configuration, atmospheric conditions and other environmental effects, the following performance with the DMI option can be achieved.

Ideal System Performance for MX60 Premium

POSITIONING POST PROCESSED ²		
No GNSS outage	Core / Pro	Premium
X, Y, Z position ¹	X, Y: <0.01 m Z: 0.01 m	X, Y: <0.01 m Z: 0.01 m
Roll and pitch	0.005°	0.0025°
Heading ³	0.015°	0.015°
60 second GNSS outage	Core / Pro	Premium
X, Y, Z position ¹	X, Y: 0.12 m Z: 0.1 m	X,Y: 0.1 m Z: 0.07 m
Roll and pitch	0.005°	0.0025°
Heading ³	0.015°	0.015°

¹ Measured in a controlled test area under Trimble conditions and procedures.

² Calculated with POSPac MMS.

³ With GAMS option, 2m baseline.

Technical Specifications

- Advanced Applanix IN-Fusion+ GNSS-Inertial integration technology
- Advanced Trimble Maxwell™ Custom GNSS survey technology with 2 × 336 tracking channels
- Primary Antenna:
 - GPS: L1 C/A, L2C, L2E, L5
 - GLONASS: L1 C/A, L2 C/A, L3 CDMA ²
 - BeiDou: B1, B1C, B2, B2A, B3³
 - Galileo¹: E1, E5A, E5B, E5AltBOC, E6²
 - IRNSS: L5

- QZSS: L1 C/A, L1 SAIF, L1C, L2C, L5, LEX
- SBAS: L1 C/A, L5
- MSS L-Band: OmniSTAR, Trimble RTX
- Secondary Antenna:
 - GPS: L1 C/A, L2C, L2E, L5
 - GLONASS: L1 C/A, L2 C/A, L3 CDMA²
 - BeiDou: B1, B1C, B2, B2A, B3³
 - Galileo¹: E1, E5A, E5B, E5AltBOC, E6² Calculated with POSPac MMS
 - IRNSS: L5
 - QZSS: L1 C/A, L1 SAIF, L1C, L2C, L5, LEX
 - SBAS: L1 C/A, L5
- High-precision multiple correlator for GNSS pseudorange measurements
- Unfiltered, unsmoothed pseudorange measurements data with low noise, low multipath error, low time domain and high dynamic response
- Very low noise GNSS carrier phase measurements with <1 mm precision in a 1 Hz bandwidth
- Data logging:
 - Time tag, status, position, attitude, velocity, track and speed, dynamics, performance metrics, raw IMU data (200 Hz), raw GNSS data (5 Hz)
- Proven Trimble low elevation tracking technology
- Real-time GNSS L1, SBAS positioning mode
- Real-time 100 Hz position, attitude output, dual IMU 200 Hz data rate logging
- Navigation output format: ASCII (NMEA-0183), binary (Trimble GSOF)
- Supported by POSPac MMS
- No export permit required
- Support for optional Distance Measurement Indicator (DMI) input
- Support for optional GNSS Azimuth Measurement System (GAMS)

¹ Developed under a License of the European Union and the European Space Agency.

² There is no official GLONASS L3CDMA or Galileo E6 ICD. The current tracking capability is based on publicly available information. Full receiver compatibility cannot be guaranteed.

³ The hardware of this product is designed for BeiDou B3 compatibility (trial version) and its firmware will be enhanced to fully support such new signal as soon as officially published ICD becomes available.

Abbreviations

The following table explains some of the acronyms and expressions used in this manual.

Abbreviation	Description
#SAT	The number of GNSS satellites being received by the GNSS receiver
Ah	Battery rating (Ampere hours)
DMI	Distance Measuring Indicator
fps	Frames per Second
GAMS	GNSS Azimuth Measurement System
Galileo	European GNSS system
GLONASS	Russian GNSS system
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
IMU	Inertial Measurement Unit
IP	Internet Protocol
LED	Light Emitting Diode
PDOP	Positional Dilution of Precision
PWR	Power
PRR	Pulse Repetition Rate
RMS	Root Mean Square
RTK	Real-Time Kinematic Positioning
SSD	Solid State Disk
SRR	Scan Repetition Rate
UTC	Universal Time Coordinated
V AC	Volts Alternating Current
V DC	Volts Direct Current
ZUPT	Zero Velocity Update

Vehicle Requirements

General

- Vehicle should be fitted with rubber wheels for use on paved roads
- No bright color to avoid exposure artefacts
- Room for system storage during transfer:
 - Dimensions of Sensor Unit transportation case: 806 × 716 × 634 mm (L × W × H)
 - Dimensions of Control Unit transportation case: 453 × 255 × 408 mm (L × W × H)
- Hatchback vehicle with upright rear door
- Minimum height of 1.60 m for the vehicle roof to comply with the minimum-distance requirement of the down-looking camera.

Mechanical

- Roof rails should withstand the required roof load
- Required dimensions of roof wings to properly install the MX SCAN Roof Rack (width × height): 85 × 30 mm
- Roof rails should cover the entire roof so the scanner position can be adjusted in case of slightly unfavorable roof rail design
- Remember the maximum lengths for system cables:
 - Power-Unit-to-Control-Unit cable: 3 meters
 - Control-Unit to-Sensor-Unit cable: 5 meters
- Allow for mounting options for Power Unit and Control Unit inside the vehicle (these two units are NOT waterproof).

Electrical

See [Technical Specifications, page 52](#) and [Installing a Power Supply in a Vehicle for Safely Operating the MX60 System, page 61](#).

- Capable of delivering the required power
- Start/stop mechanism must be switched off
- Sufficient alternator power (or additional battery pack)

Typical Installation Example



Installing a Power Supply in a Vehicle for Safely Operating the MX60 System

To safely operate the MX60 system in a vehicle, a reliable 12 V DC power supply should be made available in the vehicle. See also [Technical Specifications, page 52](#).

Typically this can be accomplished using the vehicle's own battery and charging system. With the help of a buffer battery, the MX60 system operation will be independent of the short-term behavior of the vehicle's electrical power management system.

CAUTION – Please observe the following recommendations:

- Any installation and integration of a power supply in a vehicle must be done by a professional car electrician service and are under customer's responsibility and control.
- All selected components (cable type, cable gauge, fuses, relay, etc.) have to be selected to be compliant with the MX60 input power requirements and with the local law and vehicle regulations.
- Please consider car-related restrictions like for example, car battery & energy management (Auto On/Off, Energy Management Emission Level Euro 6 and higher, etc.) as well as any restrictions given by local law.

The power input of the MX60 system should meet the following electrical requirements:

Input Voltage Range		12 to 16 V DC
DC Current	At start-up:	25 A @ 12.8 V (320 W)
	Steady-state operation:	12 A @ 12.8 V (160 W)
Battery Capacity		60 Ah min.

Typical Power Supply Setups

The two setups described below are designed with a relay, which only provides power to the connector when the alternator—and in this case the car engine—is running. This prevents the primary battery (i.e. the car battery) from going too low.

Direct connection

(Only the car battery is used.)

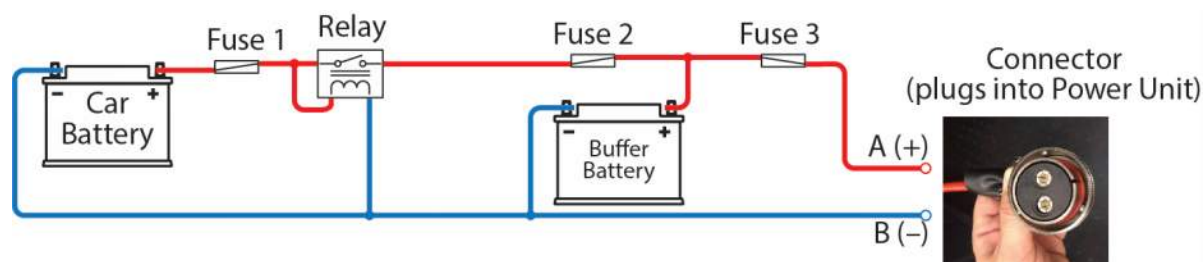
In this scenario, the provided Source-to-Power-Unit cable (P/N ; 5 meters long), with its open end, can be connected directly to the fuse and to the relay. Both the fuse and the relay should be placed close to the battery, as shown in the electrical diagram below. The current rating of fuse 3 should be 35 A.



Using a Buffer battery

(A buffer battery is used, in addition to the car battery.)

In this scenario, additional fuses (Fuse 1 and Fuse 2) are required as the buffer battery is also a power source. Fuse 1 should be placed as close as possible to the car battery and fuse 2 as close as possible to the buffer battery. As shown in the electrical diagram below, the provided Source-to-Power-Unit cable (P/N) is then connected to Fuse 3 and to the buffer battery.



For more details, please contact Trimble Support.

External-Signal Connector

This connector provides the possibility to time-synchronize other systems with the GNSS clock of the MX60 system (PPS and serial interface). Furthermore, it provides one independent trigger signal and one independent event signal.

The use of this connector requires very specific and technical knowledge. Consult the supplier of the external device you want to connect to your Trimble MX60 system to ensure proper interfacing. Based on your physical installation and interface requirements, you have to create your own cable(s) and potentially needed converter(s) of electrical signals.

System Socket: LEMO EEG.2K.314.CLL

Possible mating connector: LEMO FGG.2K.314.CLAC85

Pin No.	Name	Type	Comment
1	Trigger 1	5 V TTL	Output positive edge logic (default)
2	GND		Ground for Trigger 1
3	Event 1	High Level: > 4.0 V DC Low Level: < 0.8 V DC	Opto-isolated Input
4	GNDI		Ground for Opto-isolated Event 1
5	Reserved		Not available
6	Reserved		Not available
7	Reserved		Not available
8	Reserved		Not available
9	COM RX	RS-232	RS-232 Receiving Channel
10	COM TX	RS-232	RS-232 Transmitting Channel
11	GND		Ground for COM
12	PPS 1	5 V TTL	Output positive edge logic
13	PPS 2	5 V TTL inverted	Output negative edge logic
14	GND		Ground for PPS 1/2

WARNING – For each signal, the dedicated ground terminal (GND) must be used.

Detailed Signal Description

- **Trigger 1:**

Provides the distance-based Trigger signal configured in the TMI software.

- **Event 1:**

- The MX60 system is capable of simultaneously marking time tag events through the External-Signal connector. These tags may be used to identify the start and/or end of data collection and to synchronize the MX60 data with data from other sensors or systems.

Events are tagged with GPS, UTC. To tag an event, a signal of (High Level: > 4.0 V DC and max. 5.5 V DC; Low Level: > 0.0 V DC and < 0.8 V DC) is input into the MX60 system.

Event time tagging occurs when the MX60 system detects a rising edge (default configuration) on an event line. The time of the edge (accurate to within one microsecond) is captured. The event time is logged for use in postprocessing.

- Guard Time: 20 ms (corresponds to max 50 Hz).

- **COM (RX/TX/GND):**

- Port settings:

Setting	Value
Interface	RS-232
Baud Rate	2400 to 115200 (9600)
Parity	None
Data Bits	8
Stop Bits	1
Flow Control	None

- NMEA Messages and Data Format:

The MX60 system outputs data on the COM port using the NMEA standard 0183 format. The supported NMEA message formats are listed below. The MX60 system has several different sentence formats available for output to third-party equipment. Any or all of the sentences are available for output.

TalkerID: GP

Message	Output
GST (NMEA-0183 message: GST)	Pseudo-range measurement noise statistics
GGA (NMEA-0183 message: GGA)	Position data, with GNSS fix, geoidal separation
HDT (NMEA-0183 message: HDT)	Heading
ZDA (NMEA-0183 message: ZDA)	Time and date
VTG (NMEA-0183 message: VTG)	Track and speed
PASHR (NMEA-0183 message: PASHR)	Attitude
PPS	PPS, UTC time, PPS time recovery. See NMEA Port PPS message format below
\$EVT1 (NMEA-0183 message: PNTL,EVT)	Timed event #1
GGK (NMEA-0183 message: PTNL,GGK)	GNSS fix
RMC (NMEA-0183 message: RMC)	NMEA Recommended Minimum Specific Navigation Data

Not an NMEA 0183 message but it is compatible with the standard.

NOTE – All real-time position and orientation message strings are expressed with respect to the Reference Body Frame.

- **PPS 1/2:**

- The MX60 system uses the one Pulse Per Second (PPS) signal from a GNSS receiver for internal timing requirements. A duplication of this signal is provided on the connector to allow external equipment to be synchronized with the GNSS time clock.



The PPS output port is an active circuit. Ensure that NO 'input signal' is connected to the PPS output port, otherwise damage may result.

- The PPS signal is a TTL level strobe (pulse width 1 ms) that occurs at a 1 Hz rate. The rising/falling edge (PPS 1 / PPS 2) of the strobe is coincident with the exact GPS second. The corresponding time message that specifies the UTC time of the PPS is available in multiple formats on the COM port.

- **NMEA Port PPS message format:**

The PPS data are output in the following ASCII NMEA format:

\$ttPPS,hhmmss.ss,dddddd,wwwwww,fff.ff,pppppp,*hh<CR><LF>

Item	Definition, Values, Units
\$ttPPS	Header with talker ID \$ttPPS tt = IN or GP
hhmmss.ss	UTC time of PPS NRG hours / minutes / seconds. decimal sec
dddddd	Day offset NRG days
wwwwww	GPS week NRG weeks
fff.ff	UTC time offset NRG seconds
pppppp	PPS count NRG N/A
*hh	Checksum Hexadecimal value
<CR><LF>	N/A (NRG) Carriage return & <CR><LF> N/A Line feed

NOTE – NRG means 'No Range Given'.

Setting the Location of the GAMS Antenna

Why Use a GAMS? The GAMS helps to speed up the initialization of the navigation system inside the MX60 system. The initialization procedure is a mission-specific task, which has to be done before every mission.

Using a GAMS, not only is the initialization time reduced but also no special driving maneuvers are necessary to complete initialization.

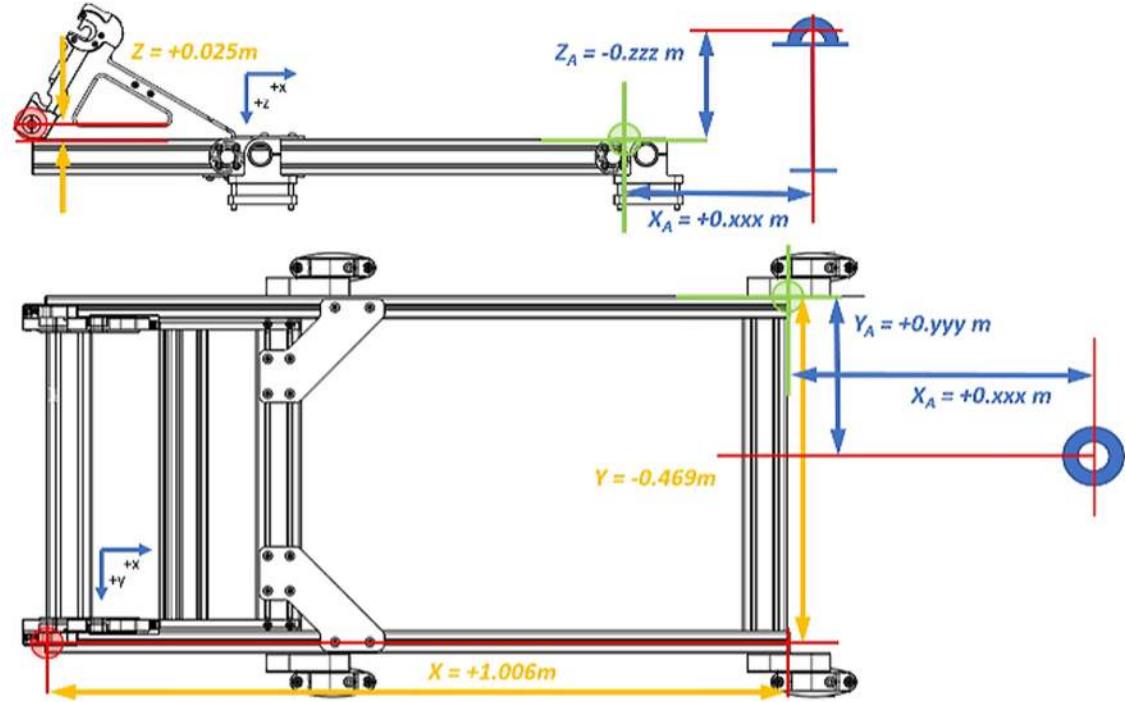
GAMS Setting Requirements. Before using a GAMS, you need to specify the three XYZ components (XYZ offsets) of the vector connecting the GAMS antenna to the External Reference Point on the Roof Rack. And remember these measurements need to be done every time you re-install the GAMS on the car roof for a new mission. Depending on how you want to use the GAMS, specific requirements need to be met when setting up the GAMS:

- **If you intend to use a GAMS just for collecting data:**
 - XYZ offsets should be measured with an accuracy of only 10 cm (or better).
 - Base length between primary and secondary antennas .
- **If you intend to use a GAMS for later being able to postprocess the collected navigation data,** tighter requirements need to be met:
 - XYZ offsets should be measured with an accuracy in the order of a few millimeters.
 - Base length between primary and secondary antennas should not be less than 2.0 m.
 - The primary and secondary antennas used **MUST** be of the same type. **DO NOT** mix different antenna types in the setup.
 - **IMPORTANT-** Offsets should always be measured with respect to the L1 antenna phase center (APC) of the corresponding antenna!

What to Measure exactly? The GAMS antenna being usually installed close to the front of the Roof Rack (on the roof of the car), it's easier to measure the offsets from the GAMS antenna to the top-left front corner of the Roof Rack (these are relatively short distances) rather than measuring those to the external reference point (these distances, which are larger, would be more difficult to measure). Because the components of the distance from the top-left front corner of the Roof Rack to the External Reference point are accurately known, it will be easy to deduce all the offset values separating the GAMS antenna from the External Reference Point.

GAMS Setup Instructions (see also illustration below):

- Measure the XYZ offsets between the GAMS antenna and the top-left front corner of the Roof Rack (this corner is shown in green below).
- Add your measured offset values (in blue) to the known offsets (in yellow). The resulting values describe the position of the GAMS antenna with respect to the External Reference Point on the Roof Rack.
- Enter these values as *Lever Arm X*, *Lever Arm Y*, *Lever Arm Z* in the TMI software when setting the GAMS before you make it active.



	Location of External Reference Point on Trimble standard Roof Rack.	
	Location of top-left front corner of the Trimble standard Roof Rack. Its offset values along the lever arms compared to the location of the External Reference Point are fixed and known. These are given in the right-hand column:	$X = +1.006\text{ m}$ $Y = -0.469\text{ m}$ $Z = +0.025\text{ m}$
	Location of GAMS antenna. Its offset values along the lever arms compared to the location of the Ext. Ref. Point are calculated as follows:	Lever Arm $X = +1.006\text{ m} + 0.000\text{ m}$ Lever Arm $Y = -0.469\text{ m} + 0.000\text{ m}$ Lever Arm $Z = +0.025\text{ m} - 0.000\text{ m}$

Where 0.xxx, 0.yyy and 0.zzz are your measured values (in meters).